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Exploring teacher acceptance of e-learning technology

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A number of studies have indicated that the successful pedagogical use of technology depends on teachers' attitudes and acceptance towards technology. In order to predict and understand teachers' technology use and acceptance, a well-defined framework is essential. The purpose of the present paper attempts to explore a model to understand teacher acceptance of e-learning technology. A self-reported questionnaire was designed to examine teacher acceptance and attitude towards an online learning platform. Data were collected from 152 in-service teachers who were studying in a part-time teacher education program in Hong Kong. The Technology Acceptance Model (TAM) was used as the core framework for analysis while additional constructs were added in order to find a better model to understand teacher acceptance of e-learning technology. A composite model including five constructs, namely, intention to use, perceived usefulness, perceived ease of use, subjective norm and computer self-efficacy, were formed and tested in the study. LISREL was used in the data analysis. It was found that subjective norm and computer self-efficacy serve as the two significant perception anchors of the fundamental constructs in TAM. However, contrary to previous literature, perceived ease of use became the sole determinant to the prediction of intention to use, while perceived usefulness was non-significant to the prediction of intention to use. Altogether, subjective norm, computer self-efficacy and perceived ease of use were able to explain 68% of the variance observed in users' intention to use the e-learning system. Implications to teacher education and teacher professional development are discussed.

Keywords: e-learning; teacher education; teacher perception; technology acceptance

Introduction

Nearly 95% of higher education institutions utilise some kind of electronic learning (e-learning) (Pollack, 2003 and some researchers even argue that e-learning is one of the most significant recent developments in the information system industry (Wang, 2003). In general, e-learning refers to asynchronous or self-paced learning (Kelly, 2001). To be more specific, e-learning refers to the use of Internet technologies to deliver a broad array of solutions that enhance knowledge and performance (Rosenberg, 2001). E-learning is based on three fundamental criteria: (1) it is networked, which makes it capable of instant updating, storage/retrieval, distribution and sharing of instruction or information; (2) it is delivered to the end-user via a computer using standard Internet technology; and (3) it focuses on the broadest view of learning – learning solutions that go beyond the traditional paradigms of training (Rosenberg, 2001, p. 28).

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The past two decades have witnessed the unprecedented growth of the Internet and an ensuing transformation in the educational landscape. The adoption of a wide variety of web-based tools has given rise to the trend of e-learning in education worldwide. Meanwhile, the increasing availability of wireless (Choi, Abbott, Arthur, & Hill, 2007) and mobile technology (Petrova, 2007) makes e-learning even more ubiquitous and pervasive. E-learning technology opens possibilities for new ways of engagement and invites innovative pedagogies. A digital mindset (Smith, 2000) is required in the teaching profession, though teachers differ widely in their attitudes and ability to cope with technology (Sutherland-Smith, 2002).

The successful use of computers in classrooms depends on the teachers' attitudes towards computers (Lawton & Gerschner, 1982). Prior studies have stated that teachers' attitudes as well as knowledge and skills in using computers are major factors affecting their initial acceptance of computer technology and their future behaviour regarding computer usage (Koohang, 1989; Violato, Mariniz & Hunter, 1989). Kluever, Lam and Hoffman (1994) also indicated that teachers' attitudes towards computers affect their instructional use of computers and likelihood of profiting from training. However, recent studies find that instructors and students are still reluctant to engage in an active or sustained manner in activities using information technology (Reffell & Whitworth, 2002). The actual formal use of information technology in undergraduate and graduate studies still remains inconsistent and varies significantly from individual courses to individual institutions (Breen et al., 2001; Marriott et al., 2004). More recent studies suggest that further research design should extend beyond the technology-based tools, for example, to include a broader range of social factors (Bielaczyc, 2006; Selwyn, 2007). Thus, in order to predict and understand teachers' technology use and acceptance a well-defined framework is essential. The present paper attempts to explore a model to understand teacher acceptance of e-learning technology and to discuss its implications to teacher education and teacher professional development.

Understanding technology acceptance

The Technology Acceptance Model (TAM) has been the most widely adopted theoretical framework to study technology acceptance. Introduced by Davis (1989), TAM was an adaptation of the "Theory of Reasoned Action" (Ajzen & Fishbein, 1980) specifically tailored for modelling user acceptance of information systems. The goal of TAM is to provide an explanation of the determinants of computer acceptance. Perceived usefulness and perceived ease of use are hypothesised to be the fundamental determinants of user acceptance (Davis, 1989). The simplified TAM was also widely validated empirically (Venkatesh, 1999; Venkatesh & Davis, 1996, 2000).

Gressard and Loyd (1985) found that the perceived usefulness of computers can influence attitudes towards computers and the amount of confidence a teacher possesses in using computers may influence his or her implementation in the classroom. Yuen and Ma (2002) found that the two independent variables 'perceived usefulness' and 'perceived ease of use' directly affect the intention to computer use as stated in the TAM. However, Legris, Ingham and Collette (2002) concluded: "TAM is a useful model, but has to be integrated into a broader one which would include variables related to both human and social change processes and to the

adoption of the innovation model". Therefore, while we took TAM as the core framework in our study, we also looked for appropriate human and social constructs to be put into the framework in order to provide a better understanding to the exploration of e-learning technology acceptance amongst teachers. After reviewing relevant literature, it is found that "computer self-efficacy" of users and "subjective norm" of the users' workplace are appropriate constructs to be considered in formulating the composite framework (e.g. Compeau & Higgins, 1995, 1999; Hall & Higgins, 2005; Ma et al., 2005; Taylor & Todd, 1995; Teo et al., 2008; Venkatesh & Davis, 1996; Venkatesh, Morris, Davis, & Davis, 2003).

The Theory of Reasoned Action is based on the assumption that human beings are usually rational and make systematic use of information available to them. Therefore, people consider the implications of their actions before they decide to engage or not to engage in a given behaviour. As most actions of social relevance are under volitional control, the theory viewed a person's intention to perform (or not to perform) a behaviour as the immediate determinant of the action. It suggests that a person would usually act in accordance with his or her intention. A person's intention is a function of two basic determinants, one personal in nature and the other reflecting social influence. The personal factor is the individual's positive or negative evaluation of performing the behaviour. This factor is termed attitude toward the behaviour which is referred to the person's judgment that performing the behaviour. The second determinant of intention was the person's perception of the social pressures put on him to perform or not perform the behaviour in question. This factor was termed subjective norm as it deals with perceived prescriptions. Generally speaking, individuals would intend to perform a behaviour when they evaluate it positively and when they believe that important others thought they should perform it (Ajzen & Fishbein, 1980).

Social cognitive theory (Compeau & Higgins, 1995; Compeau, Higgin & Huff, 1999) put forward the concept of self-efficacy, defined as the beliefs about one's ability to perform a specific behaviour, in the view that our expectations of positive outcomes of a behaviour would be meaningless if we doubted our capability to successfully execute the behaviour in the first place. The theory built on the beliefs that capability to use technology successfully is related to the decisions about whether and how much to use technology and the degree to which an individual is able to learn from training. Moreover, social cognitive theory explicitly acknowledges the existence of a continuous reciprocal interaction between the environment in which an individual operates his or her cognitive perceptions (self-efficacy and outcome expectations) and behaviour (Bandura, 1986). Thus, self-efficacy is viewed in social cognitive theory as an antecedent to use, but successful interactions with technology are also viewed as influences on self-efficacy.

Model framework and hypotheses development

While taking TAM as the core model, subjective norm and computer self-efficacy were introduced to form a composite model to explore teacher acceptance of e-learning technology. The direct and indirect effects of each construct constituted the hypotheses (Figure 1) and were tested through empirical data. Formulation of each hypothesis is described as follows.

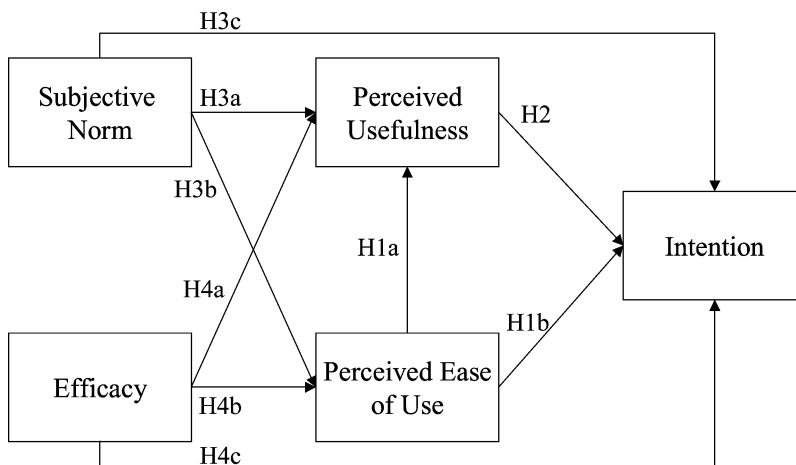


Figure 1. Model testing framework.

Perceived ease of use

Perceived ease of use refers to the degree to which the prospective user expected the target system to be free of effort. This was hypothesised to be a fundamental determinant of intention and, hence, behaviour of user acceptance to a specific application system. As validated from extensive previous studies, we propose:

- H1a: An individual's perceived ease of use has a positive effect on his or her perceived usefulness of e-learning technology.
- H1b: An individual's perceived ease of use has a positive effect on his or her intention to accept e-learning technology.

Perceived usefulness

Perceived usefulness is defined as the prospective user's subjective probability that using a specific application system would increase his or her job performance within an organisational context. This was hypothesised to be a fundamental determinant of intention and, hence, behaviour of user acceptance of a specific application system. Thus, perceived usefulness as being the fundamental construct of TAM, we propose:

- H2: An individual's perceived usefulness has a positive effect on his or her intention to accept e-learning technology.

Subjective norm

A person's subjective norm is his or her perception that most people who are important to him or her think he or she should or should not perform the behaviour in question (Ajzen & Fishbein, 1980). An individual perceives that the more others who are important to him think he or she should perform a behaviour; the more he or she will intend to do so. Thus, we propose:

- H3c: An individual's perception on subjective norm has a positive effect on his or her intention to accept e-learning technology.

We argue here that the introduction of an e-learning platform is in fact a combined effort of both the instructor and the organization. The learner will naturally perceive that the instructor would think all individual learners should use the system. At the same time, because the instructor and the organization have spent so much effort in providing the e-learning materials on the platform, the individual will also view the system as useful – otherwise the instructor would not spend so much time and effort on the it. From the learner's view, the instructor and the organization are important others in the learning community. Thus, we propose:

- H3a: An individual's perception on subjective norm has a positive effect on his or her perceived usefulness about e-learning technology.

The learner perceives the important others would think he or she should use the e-learning system. He or she would also have a general perception that, as the important others think that he or she should use it, it must be easy to use. Therefore, we propose:

- H3b: An individual's perception on subjective norm has a direct positive effect on his or her perceived ease of use about e-learning technology.

Computer self-efficacy

Computer self-efficacy is defined as the judgment of one's capability to use computer technology. The strength of a computer self-efficacy judgment refers to the level of confidence an individual has regarding his or her ability to perform various tasks in the system. Here, we argue that an individual will have a higher tendency to use the system if he or she perceives him or herself as capable or competent. Thus, we propose:

- H4c: An individual's general computer self-efficacy has a positive effect on his or her intention to accept e-learning technology.

On the other hand, if an individual views him or herself as less competent at using the system, there will be more hesitation to use the system. In general, a person will find any new systems to be difficult to use, if he or she has a low self-efficacy in using a computer. We propose:

- H4b: An individual's general computer self-efficacy has a positive effect on his or her perceived ease of use about e-learning technology.

The more the individual uses the system, the more the benefits the individual will get from the system. Just the opposite, if the individual seldom uses the e-learning platform, the individual will not be aware of the potentials and functionality of the e-learning platform. At the same time, the individual will find the system less useful. That is, a person with a high self-efficacy to computers in general will perceive a system more useful; a person with a low self-efficacy to computers in general will perceive a system less useful. Thus:

- H4a: An individual's general computer self-efficacy has a positive effect on his or her perceived usefulness about e-learning technology.

Methods

E-learning technology

This study was based on a web-based course support platform named Interactive Learning Network (ILN). Interactive Learning Network is a community-oriented

learning management system developed by a research centre of a university in Hong Kong. It is an e-learning system that allows users to interact through synchronous and asynchronous communication. Built-in features include announcement, resource, task, forum, calendar, chart, quiz and evaluation. Instructors can customise features on his other community. All learners can post and read messages in the “forum”, a place for open discussion. Instructors provide all the learning materials under “resource” for class presentation and for off-the-class download by the learners. Instructors also can create online quizzes for assessment.

Data collection

The study targeted a group of in-service teachers studying in a part-time Postgraduate Diploma in Education program offered by a university in Hong Kong. Data were collected using a user-reported self-assessment approach, deemed appropriate because of considerable literature support for its use in intention-based studies (Davis, 1989; Sheppard et al., 1988; Warshaw & Davis, 1985). At the beginning of a semester, a total of 280 questionnaires were distributed through the class representatives. Subjects were asked to return the completed questionnaires to their class representatives within one week. Class representatives collected the questionnaires sealed in an envelope and returned them to the researcher. The authors confirm that the relevant ethics approvals have been obtained.

Instrument

We adopted measurements validated by prior research studies, with wordings revised for the targeted respondents and technology context. Specifically, the questionnaire was designed to include two items for intention to use (ITU1–2), five items for perceived usefulness (PU1–5), four items for perceived ease of use (PEOU1–4), two items for subjective norm (SN1–2) and six items for computer self-efficacy (SE1–6). The particular items for each construct were listed in the Appendix, together with their sources of reference. All items were measured in a 7-point Likert scale, with 1 as “strongly disagree” and 7 as “strongly agree”, except the construct computer self-efficacy, which was measured in 10-point Likert scale. Subjects were also asked to report their usage for the past week. The degree of current usage of computer was measured in a 7-point Likert scale. Subjects were also asked to provide their background information in the first part of the questionnaire, including gender, age range, access to e-learning platform network outside lecture and whether they had had any previous formal computer training experience.

Results

Background statistics

Among the 280 distributed questionnaires, 152 questionnaires were collected (response rate about 54%). The background of the respondents is summarised in Table 1.

Regarding the “highest level of computer competence attained”, the Education Bureau of the Hong Kong Special Administrative Region Government indicated four levels of teacher information technology competence since 1998–2003, namely, Basic Level (attending 18 hours of training), Intermediate Level (attending 30 hours

Table 1. Background of respondents.

Gender	%
Male	40.8
Female	59.2
Age	
<25	3.4
25–29	38.1
30–35	27.2
>35	31.3
Access to online teaching network after face-to-face lecture	
Yes	96.6
No	3.4
Highest level of computer competence attained	
Basic	8.8
Intermediate	49.3
Upper Intermediate	37.2
Advanced	4.7

Note: $N=152$.

of training), Upper Intermediate Level (attending 30 hours of training) and Advanced Level (attending 120 hours of training). These levels were assessed according to the number of hours of training and a set of required skills achieved.

Measurement items

The descriptive statistics of the measurement items were presented with respect to mean values and standard deviations. As shown in Table 2, all the items showed generally positive perceptions towards use of the ILN, with all mean scores over 50% of the total score. All constructs were satisfied or close to the criteria of reliability (alpha coefficients ranged from 0.6875 to 0.8670). However, “intention to use” obtained a marginal alpha value of 0.5876 and would be a limitation to the final results.

We assessed the instrument’s convergent and discriminant validity by a principal component analysis extraction method using varimax with Kaiser normalisation rotation. Five components were extracted where measurement items load highly on the respective constructs than on other components (e.g. all loadings exceeding 0.6). Table 3 shows the components, corresponding measurement items and factor loadings. The components generated confirmed the corresponding constructs as hypothesised in the model framework. The factor loadings of individual items range from 0.62–0.82. The percentage of variance explained by the components ranged from 5.29–32.10 and Eigen values ranged from 1.01–6.10 (Table 3).

Overall model

The goal of path analysis is to test a “structural” model, that is, a model comprising theoretically-based statements of relationships among constructs (Kelloway, 1998). The true power of structural equation modeling is the ability to estimate a complete model incorporating both measurement and structural considerations.

Table 2. Descriptive analysis.

	Minimum	Maximum	Mean	SD	Alpha
Computer Self-efficacy (SE)					
SE1	1	10	6.18	1.88	0.8670
SE2	1	10	6.51	1.95	
SE3	1	10	6.65	1.66	
SE4	1	10	6.48	1.83	
SE5	1	10	6.85	2.17	
SE6	1	10	6.21	1.96	
Perceived Usefulness (PU)					
PU1	1	7	4.78	1.38	0.8668
PU2	1	7	4.73	1.32	
PU3	1	7	5.11	1.13	
PU4	1	7	4.56	1.37	
PU5	1	7	5.10	1.29	
Perceived Ease of Use (PEOU)					
PEOU1	1	7	5.80	1.15	0.7734
PEOU2	2	7	5.24	1.20	
PEOU3	2	7	5.17	1.06	
PEOU4	1	7	5.46	1.09	
Intention to Use (ITU)					
ITU1	1	7	5.34	1.14	0.5876
ITU2	2	7	5.21	1.25	
Subjective Norm (SN)					
SN1	1	7	4.71	1.12	0.6875
SN2	1	7	4.55	1.04	

We examined the model using LISREL 8.3 and followed suggestions by prior literature to assess the strength of each causal path by examining its direct and total effect (Bollen, 1989). There were two components in the model. First, the structural model specifies the predictive relationships among the latent variables, as shown in Figure 2. Second, the measurement model defines how the latent variables were measured (i.e. represented by indicators). Intention to use was measured by two indicators; perceived usefulness was measured by five indicators; perceived ease of use was measured by four indicators; subjective norm was measured by two indicators; and computer self-efficacy was measured by six indicators, as shown in the Appendix. This also satisfied the two/three-indicator rule that one should strive for at least two or three indicators for every latent variable (Bollen, 1989).

All model tests were on the covariance matrix and used maximum likelihood estimation as implemented in LISREL (Jöreskog & Sörbom, 1996). Fit indices for the model were considered. Overall, it was concluded that the model was suitable and valid. Standardised parameter estimates for the model are presented in Figure 2.

As shown, intention to use is predicted by perceived ease of use ($\beta = .39, p < 0.01$), which in turn was predicted by computer self-efficacy ($\beta = .30, p < 0.01$). On the other hand, perceived usefulness was predicted by both perceived ease of use ($\beta = .22, p < 0.05$) and subjective norm ($\beta = .54, p < 0.001$). The model explained 68% of the variance in intention to use, 42% of the variance in perceived usefulness and 32% of

Table 3. Factor analysis.

	Components				
	1	2	3	4	5
Computer Self-efficacy (SE)					
SE1	0.75	0.01	0.13	0.16	-0.03
SE2	0.82	0.19	0.19	-0.05	0.04
SE3	0.68	0.06	0.00	0.26	0.15
SE4	0.62	-0.16	0.26	0.05	0.36
SE5	0.77	0.16	-0.05	0.10	0.04
SE6	0.80	0.01	0.15	0.07	0.06
Perceived Usefulness (PU)					
PU1	-0.04	0.82	0.10	0.11	0.09
PU2	-0.03	0.81	0.26	-0.01	0.03
PU3	0.10	0.71	0.02	0.29	0.21
PU4	0.08	0.83	0.02	0.04	0.16
PU5	0.26	0.75	0.10	0.23	0.08
Perceived Ease of Use (PEOU)					
PEOU1	0.01	0.07	0.63	0.53	0.00
PEOU2	0.24	0.10	0.77	-0.02	0.19
PEOU3	0.06	0.07	0.72	0.11	0.21
PEOU4	0.20	0.27	0.69	0.24	-0.11
Intention to Use (ITU)					
ITU1	0.18	0.21	0.14	0.74	0.23
ITU2	0.23	0.20	0.18	0.67	0.04
Subjective Norm (SN)					
SN1	0.22	0.25	0.02	0.27	0.74
SN2	0.06	0.29	0.29	0.03	0.77
Eigenvalues	6.10	2.83	1.73	1.08	1.01
Variance explained (%)	32.10	46.98	56.07	61.77	67.06

the variance in perceived ease of use. A summary of all the parameter estimates and the *t*-values are presented in Table 4.

Discussion

From the resulting model (Figure 3), it was found that perceived usefulness was not significant to intention to future use of the e-learning system. This is not consistent to previous studies using TAM in predicting intention and usage (Venkatesh et al., 2003). Perceived ease of use was found significant and powerful in predicting intention to use. Perceived ease of use had a significant direct positive effect on users' intention to use the e-learning system with standardised path coefficient being 0.39. This coefficient suggested that every unit increment in perceived ease of use would strengthen an individual's (positive) intention to use an e-learning system. Perceived ease of use had been found significant in most previous studies. On the contrary, perceived ease of use was much stronger. It was even the sole and dominant determinant in the model in predicting intention to use. This seems to indicate that the perceived ease of use amongst teachers is extremely important.

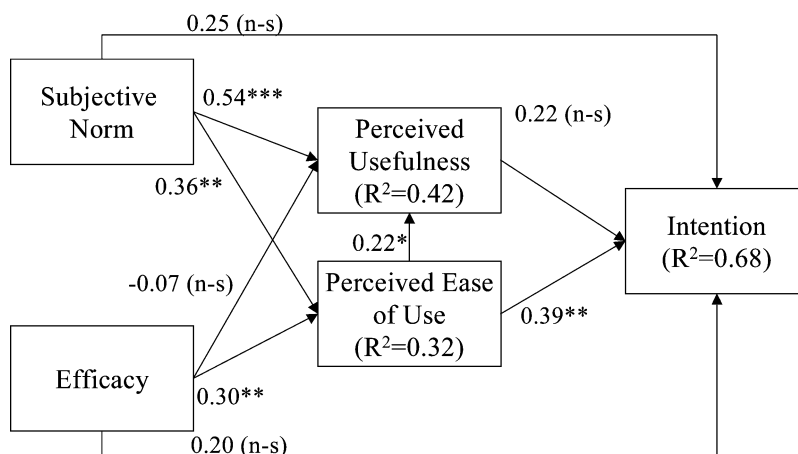


Figure 2. Model testing results.

Table 4. Summary of hypotheses testing.

Hypotheses	Path	Path Coefficients	<i>t</i> -values	Results
H1a	SN→PU	0.54***	4.69	Supported
H1b	SN→PEOU	0.36**	3.20	Supported
H1c	SN→ITU	0.25	1.74	Not Supported
H2a	SE→PU	-0.07	-0.76	Not Supported
H2b	SE→PEOU	0.42***	2.87	Supported
H2c	SE→ITU	0.20	1.92	Not Supported
H3a	PEOU→ITU	0.39**	3.23	Supported
H3b	PEOU→PU	0.22*	2.09	Supported
H4	PU→ITU	0.22	1.84	Not Supported

Note: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Subjective norm was found not significant in predicting intention to use. However, it is highly correlated with both perceived usefulness and perceived ease of use. The standardised path coefficients are 0.54 ($p < 0.001$) and 0.36 ($p < 0.01$) towards perceived usefulness and perceived ease of use, respectively. These significant correlations provide us implications to the understanding of the perception formation of users. It is also important to note that social influence in technology adoption is crucial amongst teachers. Computer self-efficacy was found not significant in predicting intention to use. However, it had a significant direct positive effect on users' perceived ease of use, with standardised path coefficient of 0.30 ($p < 0.01$). This was consistent with previous results of computer self-efficacy being an antecedent of perceived ease of use in explanation of the two constructs (Taylor & Todd, 1995; Venkatesh & Davis, 1996). Furthermore, subjective norm and computer self-efficacy serve as the two significant perception anchors of the fundamental constructs in TAM.

The explanatory power of the model for individual construct was examined using the resulting R square value for intention to use. Together, subjective norm, computer self-efficacy and perceived ease of use were able to explain 68% of the

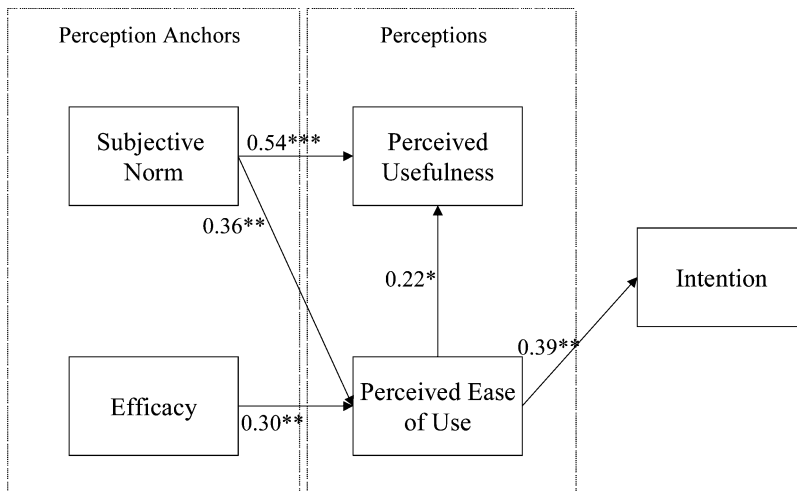


Figure 3. E-learning technology acceptance model.

variances observed in users' intention to use the e-learning system. Compared with previous TAM studies, the resulting explanatory power was comparable and strong. Szajna (1996), Taylor and Todd (1995) and Mathieson (1991) reported 52, 52 and 70%, respectively. The comparable and strong R square value of this study suggested a good strength of the model under investigation. Thus, the model is acceptable.

What does this model tell us about teacher education and teacher professional development? Firstly, subjective norm was an important and significant determinant to perceived ease of use and, hence, intention to use. Thus, it seems important to foster a school culture such that the "important figures" could provide a model to the use of e-learning technology. It would help teachers to form the perception that the important figures of the school would like them to use technology. Who are the important figures? It varies amongst organisations. In the school context, principals or head teachers are probably one of the prominent figures in promoting the use of e-learning technology. However, it remains a question about the role of school leaders in teacher professional development.

Secondly, computer self-efficacy was another important and significant determinant to perceived ease of use and, hence, intention to use e-learning technology. It is important to build up teachers' confidence in using technology in general, which would, hence, increase their willingness to use other e-learning technology in the future, as the development of teachers' positive attitudes towards technology is considered to be a key factor in fostering technology integration in teaching and learning (Yuen, Law & Chan, 1999). Summers (1990) found that lack of knowledge and experience in the computing area is one of the most common reasons for teachers' negative attitudes towards computers. Therefore, it is useful to develop a school environment that enables teachers to have more hands-on experience in new technologies (Allan, 2007; Fleming, Motamedi, & May, 2007) as Yunus (2007) indicated that teachers' development necessitates not merely providing additional training opportunities, but also aiding them in experimenting with technology before being able to use it in their classroom.

Thirdly, while perceived usefulness was found not significant in the model, we need to rethink whether the benefits and functionality of e-learning technology

should be emphasised in the first place in teacher training. More “useful” functionalities would only be introduced after a steady state of implementation, in particular the teachers are familiar with the e-learning technology they are using.

While acknowledging the implications to teacher education and teacher development in e-learning technology of the resulting model, we would also like to address the issue of applicability of TAM in examining e-learning technology. In this small-scale study we have investigated the perceptions of teachers in using an e-learning system, which aimed to enhance teaching and learning in their studies. Although the results from the data analysis were generally good, the low alpha value of the intention to use might hinder the validity of the final model. The small sample size and low alpha value are obvious limitations to the present study. However, the model-fit of the final model was close to most of the prior studies referenced. We argue the final model gives a better understanding of teacher acceptance of e-learning technology and raises a number of fundamental issues of technology integration concerning the preparation of teachers for new ways of engagement in the twenty-first century as stressed in literature such as Hunt (1997). These issues merit further investigation.

Notes on contributors

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Appendix 1.

Measurement items	
Intention to Use (ITU), adapted from Fishbein and Ajzen (1975)	
ITU1	I intend to use Interactive Learning Network (ILN) when it becomes available in my school and at home.
ITU2	I intend to use ILN in my study as often as possible.
Perceived Usefulness (PU), adapted from Davis (1989)	
PU1	Using ILN improves my performance.
PU2	ILN enables me to accomplish tasks more quickly.
PU3	Using ILN enhances my effectiveness on the study.
PU4	Using ILN increases my productivity.
PU5	Overall, I find ILN useful in my study.
Perceived Ease of Use (PEOU), adapted from Davis (1989)	
PEOU1	Learning to operate ILN is easy for me.
PEOU2	It is easy for me to become skillful in using ILN.
PEOU3	My interaction with ILN is clear and understandable.
PEOU4	Overall, I find ILN easy to use.
Subjective Norm (SN), adapted from Fishbein and Ajzen (1975)	
SN1	Those people who are important to me would strongly support my using ILN in my study of the course.
SN2	People whose opinions I value would prefer me to use ILN in my study of the course.
Computer Self-efficacy (SE), adapted from Compeau and Higgins (1999)	
SE1	I could complete the job using ILN if I had only the software manuals for reference.
SE2	I could complete the job using ILN if I had seen someone else using it before trying it myself.
SE3	I could complete the job using ILN if I had a lot of time to complete the job for which the ILN was provided.
SE4	I could complete the job using ILN if I had just the built-in help facility for assistance.
SE5	I could complete the job using ILN if someone showed me how to do it first.
SE6	I could complete the job using ILN if I had used a similar package before this one to do the same job.
